EX-05

BIRLA VISHVAKARMA MAHAVIDYALAYA ENGINEERING COLLEGE
(AN AUTONOMOUS INSTITUTION)
DEPARTMENT: ELECTRICAL ENGINEERING
4EE34 :- EXTERNAL PROJECT

Title: Power System Studies of Distribution System Using ETAP

Abstract

Heavy-duty mineral processing plants demand continuous electricity to prevent costly production downtime. This project utilizes ETAP software to execute a comprehensive electrical examination of an industrial distribution network. The in-depth study includes Load Flow and Short Circuit studies to strictly verify equipment ratings, alongside precise Protective Device Coordination and Arc Flash hazard assessments. Crucially, the initial analysis uncovered substantial personnel safety weaknesses. By optimizing protective relay coordination and specifically integrating arc flash detection relays, the network's Fault Clearing Time was successfully lowered from 0.804 seconds to 0.460 seconds. This essential modification successfully reduced the maximum Arc Flash Incident Energy from a hazardous 47.27 cal/cm² (DANGER category) to 27.76 cal/cm² (Level 4). Ultimately, these results guarantee strict adherence to NFPA 70E vital workplace safety regulations.

Introduction

Mineral processing plants require a constant source of electricity to operate heavy, high-inertia machinery such as grinding mills and conveyors. In these harsh industrial conditions, electrical instability is quite costly. A single "nuisance trip" or extreme voltage sag during a large motor beginning can shut down a production line, resulting in serious mechanical blockages and enormous financial losses due to operational downtime. Furthermore, handling the massive fault currents and limiting arc flash dangers in these plants is vital to crew safety. This study addresses these specific difficulties by examining a mineral plant's distribution network to assure system stability, safety, and continuous production.

System Overview

- **Grid Supply:** Power is received at 132 kV from the utility grid and stepped down to 11 kV via two 25/31.5 MVA transformers.
- **Captive Power:** The network integrates a Waste Heat Recovery System (WHRS) Generator supplying 12 MW at 11 kV.
- **Distribution:** Power is further reduced to 415V to supply secondary motor loads and control centers.

Methodology

- **Load Flow Study (IEEE 3002.2):** Evaluated active and reactive power to optimize transformer tap settings and ensure bus voltages remain stable within a ±5% tolerance during peak production.
- **Short Circuit Study (IEC 60909):** Calculated maximum fault currents to verify the interrupting ratings of circuit breakers, switchgears, cables , etc. (Device Duty) and evaluated minimum fault currents for relay settings.
- **Relay Coordination (IEEE-242):**Relay coordination is the strategic process of setting protective devices in a power system to ensure the relay closest to a fault operates first. This selective tripping isolates only the damaged section, preventing widespread outages and protecting expensive equipment. Backup relays provide secondary protection if the primary system fails.
- **Arc Flash Assessment (IEEE 1584 & NFPA 70E):** Computed incident energy levels to determine safe working distances and assign proper Personal Protective Equipment (PPE) categories.

Block Diagram

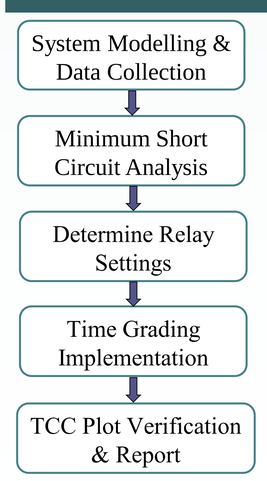


Fig 1. Relay Co-Ordination

Equipment Safety

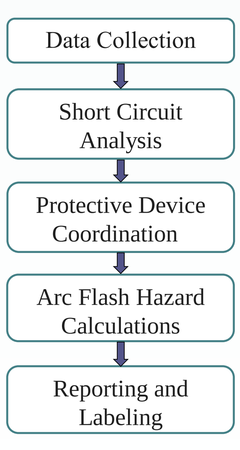


Fig 2. Arc Flash Analysis

Human Safety

Power Distribution Layout

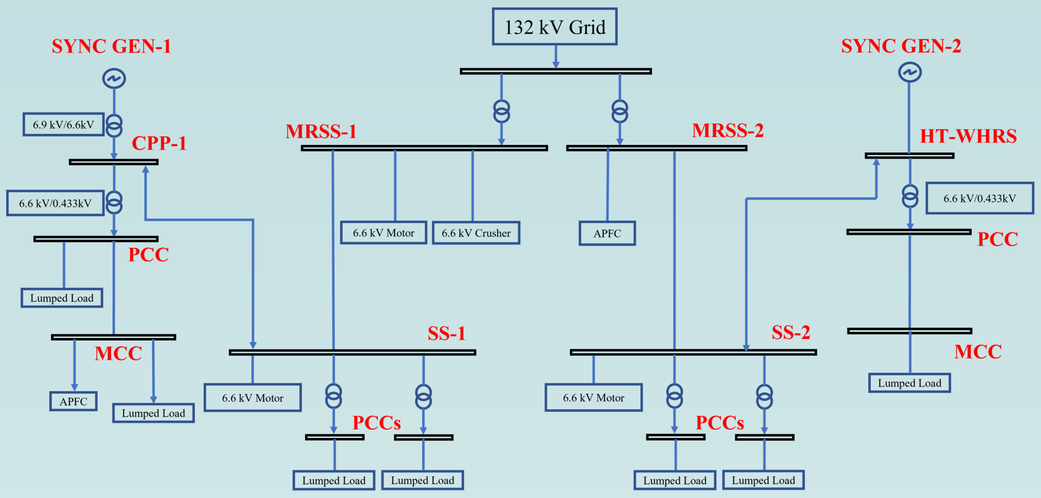


Fig 3. Power Distribution Diagram

Relay Co-Ordination

Plug Setting = Full Load Ampere / CT Ratio

Plug Setting Multiplier = Fault Current / (CT Ratio * Plug Setting)

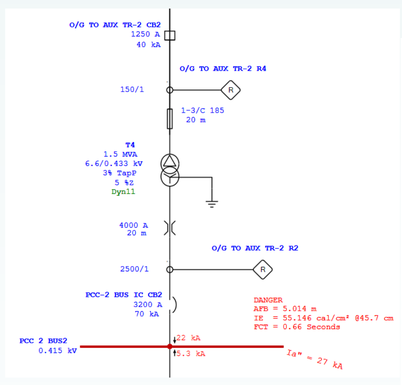
Relay Operating Time (At TMS=1) = $\frac{\beta}{(PSM)^{\alpha} - 1}$

Time Multiplier Setting = Selected Time of operation / Time of operation at PSM

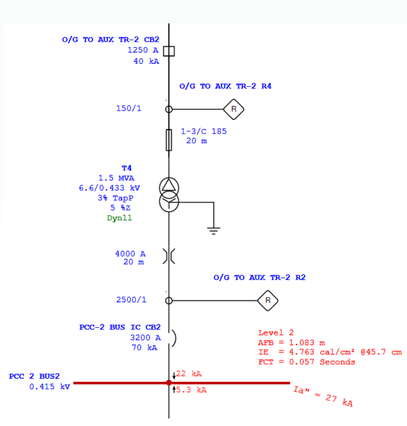
Curve Type	β	α
Standard Inverse	0.140	0.020
Very Inverse	13.5	1
Extremely Inverse	80	2
Long Time Inverse	120	1

Type of Feeder	Phase Fault IDMT Pick up	Phase Fault DMT Pick up	Earth Fault Pick up
Transformer Feeder	> 120% of Transformer FLA + Starting current of Largest motor on downstream network	Maximum out of following, >8 x Full Load Current >1.3 x maximum LT through Fault current	For NGR Earthed: 10%-30 of NGR rating For Solidly Earthed: 20-50% of fault Current
Plain/Tie/Incomer Feeder	> 110-120% of maximum feeder loading + starting current of largest motor of downstream network	< 67% of through fault current	
Motor Feeder	> 120-130 % of Motor Full Load Current	Motor starting current < set value < 67% of through fault current	

Arc Flash Results



With Relay Co-Ordination



With Arc – Flash Mitigation

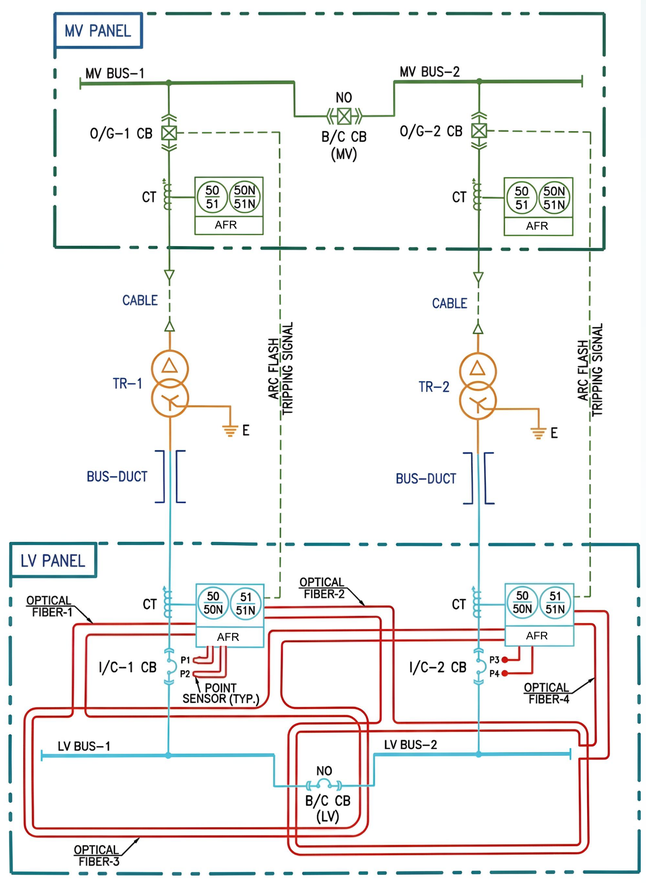


Fig 4. Arc Flash Detection Relay Scheme

Boundary Marking

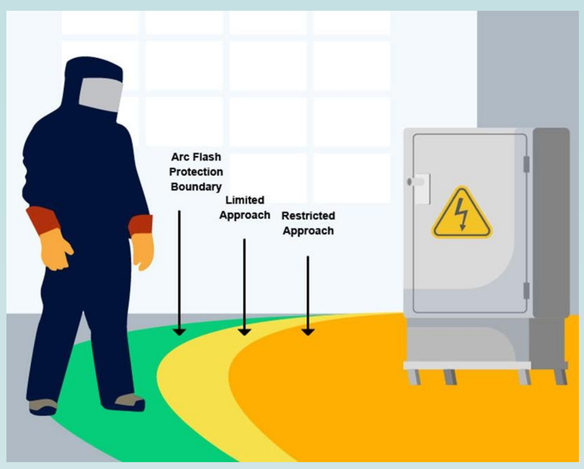


Fig 5. Arc Flash Boundary Marking

TCC Graph

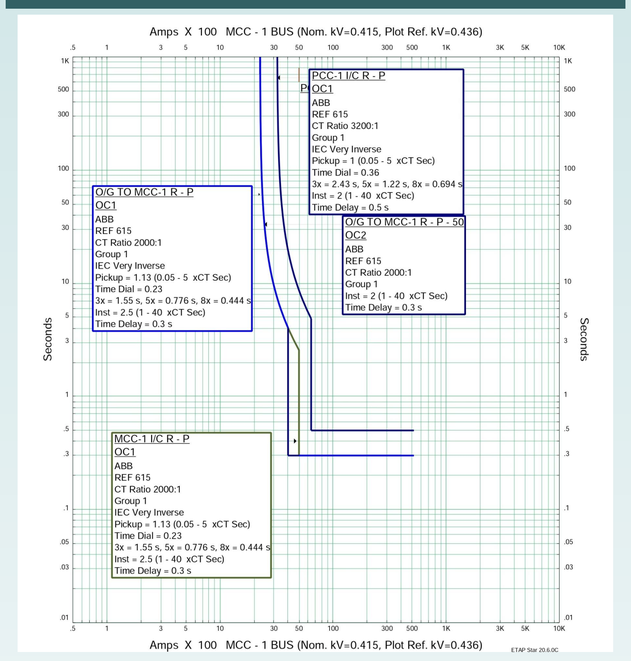


Fig 6. TCC Curve

Conclusion

Through comprehensive ETAP analysis, this project successfully optimized the industrial distribution network's reliability and safety. By refining relay coordination and integrating Arc Detection Relays, fault clearing times and dangerous incident energy levels were drastically reduced. These crucial mitigation strategies ensured strict adherence to NFPA 70E standards, effectively protecting personnel while maintaining continuous plant production.

References

1. Power System Protection and Switchgear'- Bhuvanesh.A.Oza, Nirmal Nair, Rashesh Mehta, Vijay Makwana
2. IEEMA Journal -Dr K Rajamani & Co Authors
3. NFPA 70E, Standard for Electrical Safety in the Workplace, National Fire
4. Protection Association.
5. IEEE Std 1584, IEEE Guide for Performing Arc-Flash Hazard Calculations.
6. IEC 60909, Short-Circuit Currents in Three-Phase A.C. Systems
7. IEEE Std 0242 2001 (Buff) Protection and Coordination

Scan here to view result SLDs

Scan for Website

Team

Guided by : **Faculty Guide: Prof. Ajay M. Patel**
Industry Guide: Siddharth R. Mistry
Name of Industry: Elcon Engineers Pvt Ltd.

Project Team :
Name : Aadil A. Vahora (ID NO. 22EE074)